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Annulation

Proline Derived Bicyclic Derivatives through Metal Catalysed Cyclisations of Allenes: Synthesis of Longamide B, Stylisine D and their Derivatives

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Abstract: Annulations of allene-substituted proline derivatives promoted by transition metals have been investigated as a general way to access bicyclic structures with a bridgehead nitrogen. Two processes, Pd- and Ag-catalysed cyclisations, have been employed complementary to control the substitution pattern of the product. The investigated methodologies afforded the bicyclic derivatives in comparable yields, while Ag-catalysis showed higher level of diastereoselectivity. Both processes have been utilized in the synthesis of naturally occurring pyrroles longamide B, stylisine D and their derivatives.

Introduction

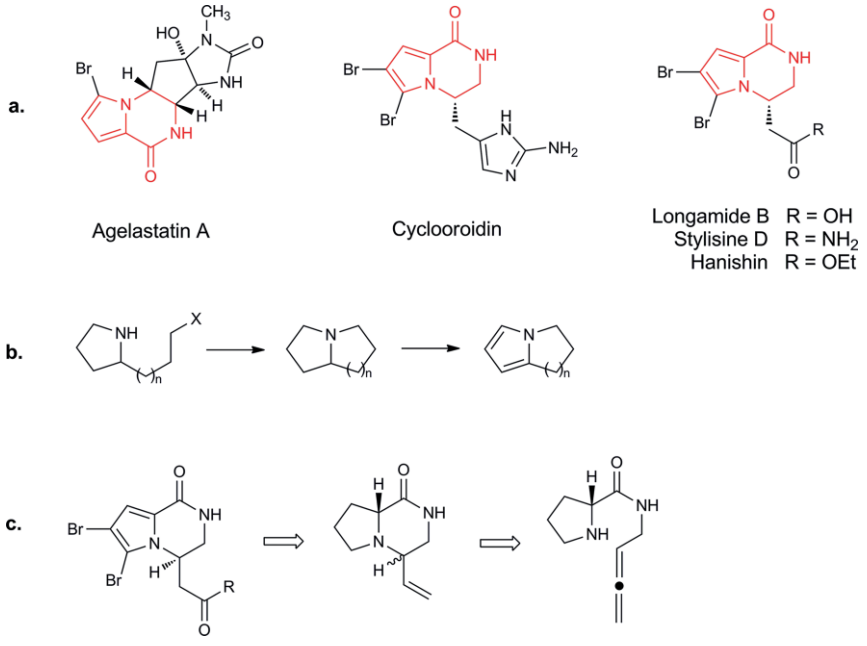
Pyrrolidine derived bicyclic skeleton, in particular one which possesses bridgehead nitrogen, is structural architecture found in many natural products and biologically active compounds.[1–6] Variations of this scaffold are numerous as indicated by large classes of pyrrolizidine and indolizidine alkaloids.[7–10] Further structural alternative, also found in naturally occurring substances, is denoted by pyrrole ring in place of pyrrolidine moiety.[11–16] Synthetic approaches to these bicyclic structures are usually based on the annulation of pyrrolidine ring, which can be accomplished by a range of methodologies (Scheme 1b).[17–25] As pyrrolidine ring can be transformed into pyrrole via oxidative processes, this approach is also valuable for the synthesis of related pyrrole derivatives, as demonstrated by the recent preparation of agelastatin.[14] Important alkaloids containing aforementioned structural motif are bromopyrroles, a large group of secondary metabolites produced by marine organisms.[26] This diverse family of alkaloids includes more than 140 members among which a number of biologically potent molecules has been uncovered such as antagonists of serotonergic receptors, kinase inhibitors, and anticancer agents (Scheme 1a).[27–29] Particularly remarkable is the antitumor activity of Agelastatin A against blood cancer cell lines. Antiprotozoal and antibacterial properties have been reported for longamide B, displaying a potent activity against African trypanosome (IC_{50} 1.53 μg/mL) and moderate activity against *Bacillus subtilis* and *Staphylococcus aureus* with minimal inhibitory concentrations MIC of 45 μg/mL and 55 μg/mL respectively.[30,31] An ethyl ester of longamide B, hanishin, displays cytotoxic activity against NSCLC-N6 human non-small-cell-lung carcinoma.[32] The structural diversity coupled with potent biological activities have sparked interest of synthetic chemists and in the last decades numerous syntheses of bromopyrrole alkaloids have been accomplished.[13,29,33–41] Bicyclic core, pyrroloketopiperazine, represents a common structural motif of a large number of bromopyrrole alkaloids and can be accessed by some of the abovementioned methods.

Our interest in the annulation of proline derivatives was initiated as potential approach to the synthesis of bromopyrrole alkaloids such as cyclooroidin and longamide B and stylisine D (Scheme 1a). Our envisaged strategy was based on the hypothesis that the piperazinone core of bromopyrroles could be accessed via intramolecular nucleophilic addition to allene moiety catalysed by transition metals as outlined in Scheme 1c. In this type of cyclization processes Pd, Au and Ag have been heavily utilized.[42–44] While Pd-catalysed reactions proceed via π-allylpalladium intermediate, Au and Ag can, due to their Lewis acidic character, coordinate to either allenic double bond allowing for an easier nucleophilic attack on this moiety.[45–47] The outcomes in regioselectivity depend on the substrate itself and oftentimes result in formation of the exocyclic double bond that can be utilized in further transformations. In case of our proposed synthesis of longamide B the secondary amine functionality of the proline ring is ideally positioned for the metal promoted cyclization onto the proximal allenyl substituent (Scheme 1c). In addition, the chirality of proline, which would be eventually sacrificed due to its oxidation to pyrrole, might promote stereoselectivity in the cyclization step, which creates a novel chiral carbon.

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Scheme 1. a. Selected bioactive bromopyrrole alkaloids; b. Annulations of pyrrolidine ring; c. Our strategy for the synthesis of pyrroloketopiperazine skeleton.

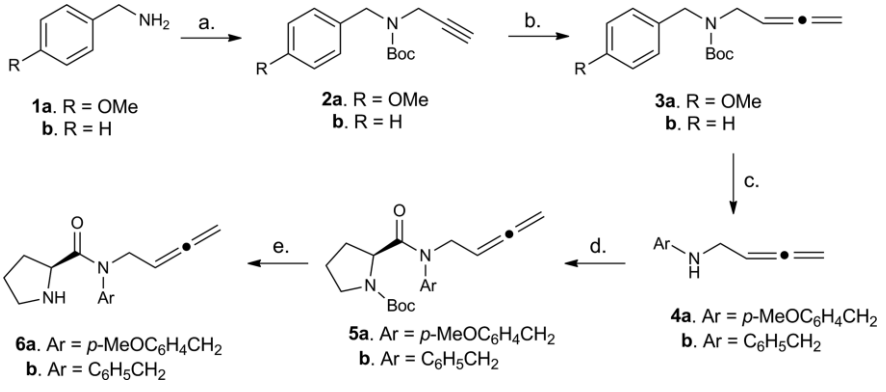
Results and Discussion

Our initial work regarding the exploration of annulation of proline ring required a synthesis of suitable allene derivative. This was shown to be a straightforward task initiated with propargylation of the protected benzylamine followed by several routine transformations to afford the key cyclization precursor proline derivative **6** (Scheme 2).[48–50]

With pyrrolidine **6** in hand we explored the cyclization process as outlined in Table 1. Various conditions based around metal-promoted transformations were screened in the reaction employing benzyl protected amide **6b**.

With catalytic amount of metal salts all reactions were very slow and needed up to five days for good conversion. Metals such as Au^{3+} and Cu^{2+} afforded products **7a**/**7b** in moderate yields (entries b and e Table 1) and with moderate diastereoselectivity. The Ag^{+} salts performed generally better (entries i–t Table 1), but, unfortunately, at the expense of quantities used to promote the reaction efficiently. Even so, the availability of Ag-salts prompted their detailed study.[51] After experimentation with additives and various quantities of metal salts the cyclization protocol was established. The reaction time was decreased to 18 h and yield was increased to 83 % with 80 mol-% of the metal salt (entry r, Table 1). Under these conditions the product was obtained as a separable mixture of **7a** and **7b** in 17:83 ratio respectively. The stereochemistry of the major component, examined by NOESY experiments, was established as *trans*. Comparison of $AgNO_3$ and AgOTf under the same conditions (entries p/s and entries q/t, Table 1) suggested that the efficiency of the cyclization process was reliant on Lewis acidity of the metal salt as better yields were generally afforded with AgOTf.

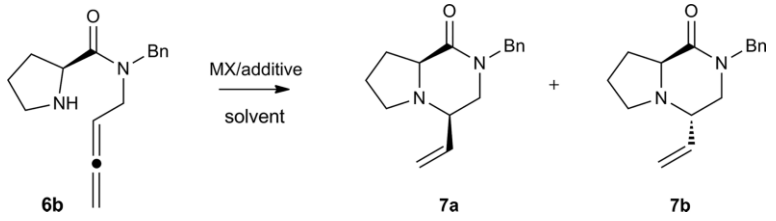
Having optimized the annulation step to access bicyclic compound **7**, longamide B was then synthesized through several additional steps (Scheme 3). Oxidation of pyrrolidine **7c**, employing MnO_2, proceeded in moderate yield to afford pyrrole **8**. Hydroboration of **8** followed by oxidation with peroxide gave the primary alcohol which was thereupon used in deprotection



Scheme 2. Synthesis of cyclization precursor **6**. a. (i) Boc_2O, NaOH, THF/H_2O, 0 °C to r.t. (ii) propargyl bromide, NaH, DMF, 0 °C to r.t., 79 %; b. $(CH_2O)_n$, iPr_2NH, CuBr, dioxane, 110 °C, 72 %; c. TFA, CH_2Cl_2, r.t., 96 %; d. Boc-L-proline, EDCI, HOBt, DIPEA, CH_2Cl_2, r.t., 88 %; e. TFA, CH_2Cl_2, r.t., 99 %.

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Table 1. Optimization of metal-catalysed cyclisation of allene-amine.



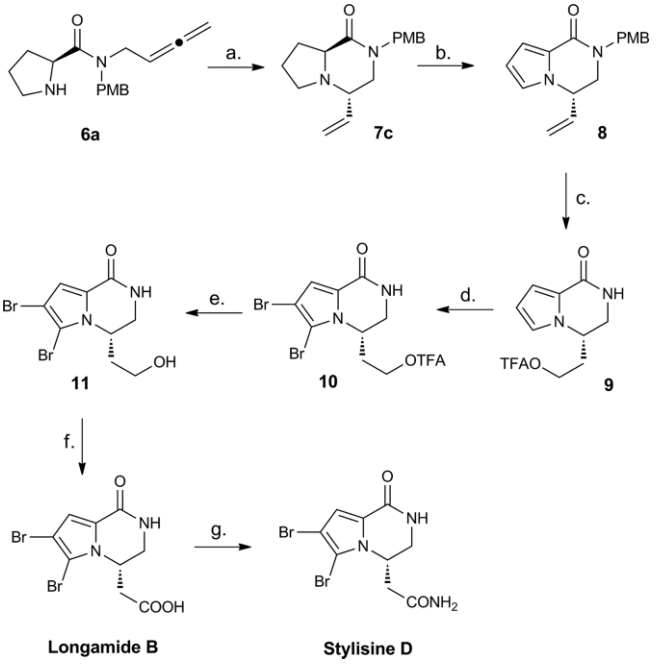
6b **7a** **7b**

Entry	MX/additives	MX (mol-%)	Solvent	Time	NMR yield [%]	Ratio (**7a**/**7b**)
a	$AuCl_3$	5	DCM	5 d	48	39:61
b	$AuCl_3/CaCO_3$/MS4Å	5	DCM + 5 % ACN	5 d	55	21:79
c	Ph_3PAuCl	5	DCM	5 d	/	/
d	$Cu(OTf)_2$	5	DCM	5 d	Trace	/
e	$Cu(OTf)_2$	5	Acetone	5 d	57	21:79
f	$Cu(OTf)_2/CaCO_3$/MS4Å	5	Acetone	5 d	Trace	/
g	$TiCl_4$	5	DCM	5 d	/	/
h	K_2PtCl_4	20	Acetone	5 d	/	/
i	Ag_2O	20	Acetone	18 h	28	29:71
j	Ag_2CO_3	20	Acetone	18 h	Trace	/
k	$AgNO_3$	20	Acetone	18 h	19	31:69
l	CH_3COOAg	20	Acetone	18 h	21	30:70
m	AgOTf	20	Acetone	3 d	/	/
n	AgOTf	20	DCM	18 h	38	26:74
o	$AgNO_3/K_2CO_3$	20	Acetone + H_2O	18 h	25	33:67
p	$AgNO_3/CaCO_3$	20	Acetone + H_2O	18 h	21	31:69
q	$AgNO_3/CaCO_3$	40	Acetone + H_2O	18 h	55	14:86
r	$AgNO_3/CaCO_3$	80	Acetone + H_2O	18 h	83[a]	17:83
s	AgOTf/$CaCO_3$	20	Acetone + H_2O	18 h	50	19:81
t	AgOTf/$CaCO_3$	40	Acetone + H_2O	18 h	74	23:77

[a] Isolated yield.

step. Our initial approach involved utilization of benzyl protection (benzyl in place of PMB in **8**), however, the cleavage of this group proved to be challenging with all attempts giving back starting materials or complex mixtures. Interestingly, cerium ammonium nitrate (CAN) attempted deprotection of N-benzyl derivative gave 5-nitropyrrole product with intact protecting group. To circumvent this problem, we utilized *p*-methoxybenzyl (PMB) protection of the amide and that group was readily cleaved with warm trifluoroacetic acid (TFA) in the presence of 2 equivalents of anisole to afford compound **9**. The deprotection step was followed by the esterification of primary alcohol functionality. This did not hinder the subsequent bromination step (Scheme 3, reaction d.), on the contrary, the ester functionality increased the overall lipophilicity of the compound allowing for an easier purification process. Thus, the bromination of **9** mediated by N-bromosuccinimide (NBS) followed by hydrolysis of trifluoroacetic ester afforded dibromopyrrole **11**. The final oxidation of the primary alcohol functionality proceeded in 83 % yield to give the targeted longamide B with spectra and optical rotation matching the ones reported in the literature ($[\alpha]_D$ = −5.5 (c 0.20, CH_3OH) in comparison to Fukase et al. ($[\alpha]_D$ = −5.6 (c 0.40, CH_3OH)).[37] Another bromopyrrole, recently isolated from marine sponge *Stylissa massa*, stylisine D, was easily synthesised by transforming the carboxylic group of longamide B into the amide.[52]

Having completed the synthetic route for the preparation of bromopyrrole natural products we next explored the possibility of applying the related Pd-promoted transformations to access

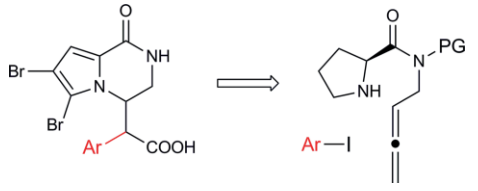


6a **7c** **8** **9** **10** **11** **Longamide B** **Stylisine D**

Scheme 3. Synthesis of longamide B and stylisine D. a. $AgNO_3$, $CaCO_3$, acetone/H_2O, r.t., 69 %; b. MnO_2, THF, 85°C, 57 %; c. (i) 9-BBN, THF, then NaOH, H_2O_2, (ii) anisole, TFA, 60°C, 55 %; d. NBS, THF, r.t., 85 %; e. NaOH, THF/H_2O, r.t., 99 %; f. TEMPO, $PhI(OAc)_2$, $NaHCO_3$, MeCN/H_2O, r.t., 83 %; g. NH_4OH, EDCI, HOBt, DIPEA, THF, r.t., 81 %.

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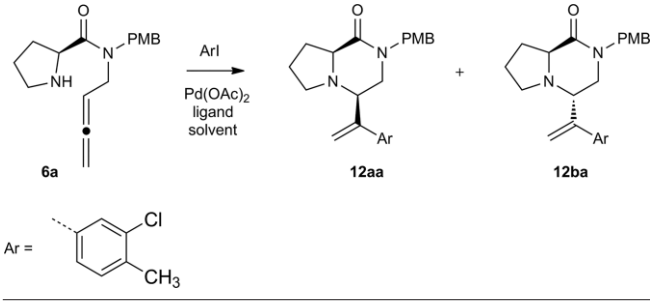
longamide B derivatives (Scheme 4). Hypothetically, carbopalladation of **6**, followed by an intramolecular nucleophilic substitution would produce analogues of this natural product with functionalization of α-carbon next to carboxylic acid. Compounds possessing substituents in this position with antifungal potential have recently been isolated from natural sources.[53]



Scheme 4. Access to α-substituted longamide derivatives.

The optimization of cyclization process of **6a** under Pd-catalysis is outlined in Table 2.[54] The reaction was performed with 1.2 equivalents of arylhalide and $Pd(OAc)_2/PPh_3$ catalytic system in DMF and in the presence of K_2CO_3 as a base. Primarily formed π-allylpalladium intermediate, obtained upon addition of arylpalladium species onto the allenic functionality, reacted intramolecularly with the secondary amine to produce bicyclic product. Unfortunately, the expected product was isolated in only 20 % yield as 1:1 mixture of diastereomers **12aa** and **12ba** (Table 2, entry a).

Table 2. Pd-catalysed cyclisation of proline derivative **6a**.



Entry[a]	Ligand	Solvent	Yield [%][b]
a	PPh_3	DMF	20
b	XantPhos	DMF	38
c	XantPhos	MeCN	58

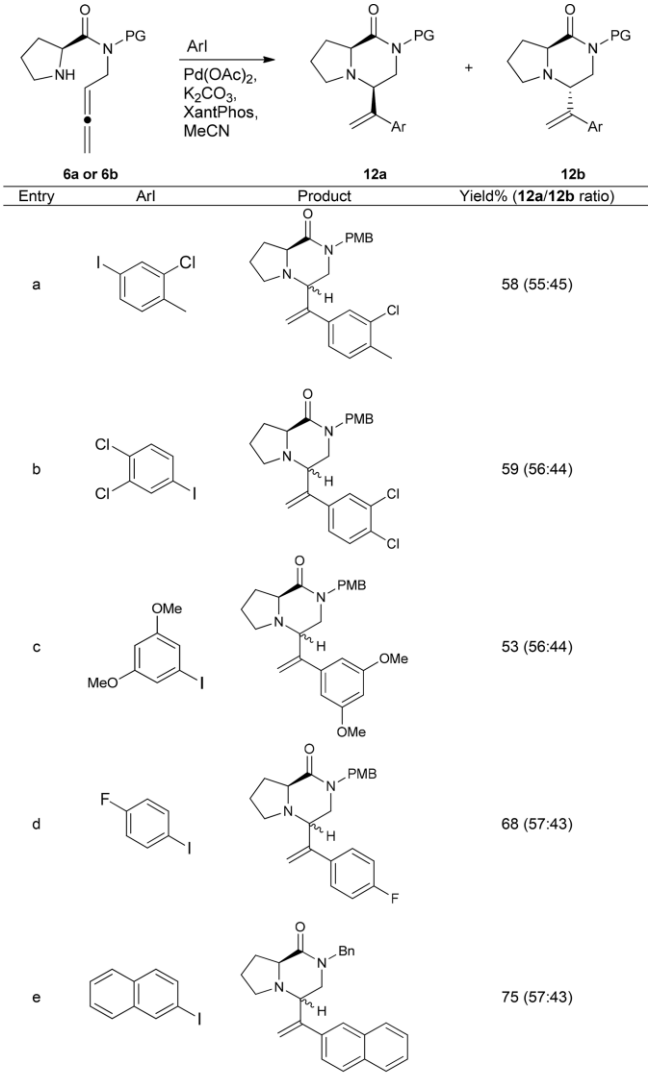
[a] Conditions: allene **6a** (0.1 mmol); ArI, (0.1 mmol); $Pd(OAc)_2$, (0.01 mmol); PPh_3 (0.02 mmol) or XantPhos, (0.01 mmol); K_2CO_3, (0.23 mmol); MeCN, (3 mL); reflux, 18 h. [b] Isolated combined yield.

When PPh_3 ligand was replaced with XantPhos the yield improved to 38 % (Table 2, entry b), while performing the reaction with XantPhos and in MeCN (Table 2, entry c) advanced the reaction yield further to 58 %. These optimized conditions were used for brief exploration of the reaction scope (Table 3). All products (Table 3, **12aa** – **12be**) were isolated in moderate to good yields but, disappointingly, with almost no diastereoselectivity.

After a brief exploration of the Pd-promoted cyclisations, we continued with transformation of the cyclic product **12ba** as already described for the preparation of parent molecule longamide B in order to demonstrate appropriateness of the developed synthetic route for the preparation of longamide derivatives. All previously utilized synthetic steps, as outlined in Scheme 5, proceeded satisfyingly. The hydroboration of **13** gave a mixture of diastereisomers with major component **14** isolated in an overall 32 % yield after the subsequent removal of PMB protecting group. The stereochemistry of newly created stereogenic centre was proposed based on the previously developed models for the hydroboration processes.[55] The final oxidation of primary alcohol, however, gave a mixture of diastereoisomers due to sensitivity of a new chiral centre now being next to a carboxylic group. At this point, we did not explore other oxidative procedures in order to reduce the extent of the isomerization process.

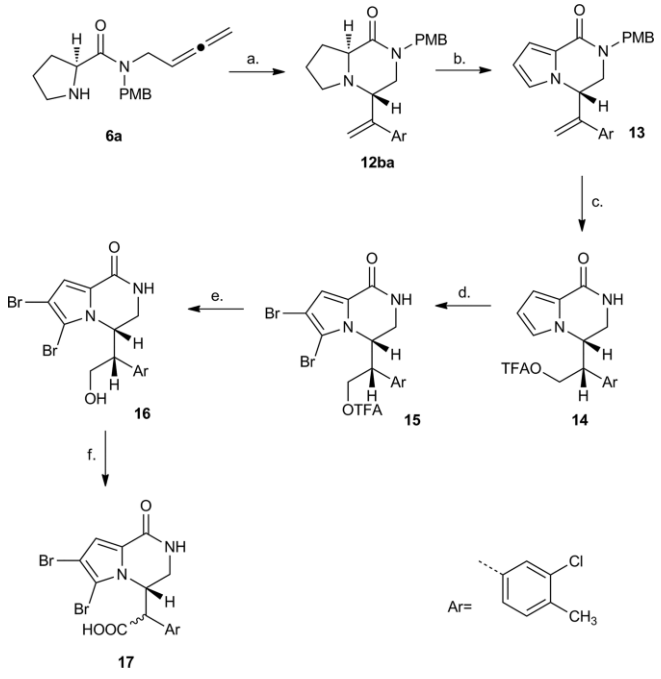
The above results demonstrated application of metal promoted annulation of pyrrolidine ring in the synthesis of bromopyrrole alkaloids but the method might have wider potential as it can be adjusted to meet structural requirements of large classes of pyrrolizidine and indolizidine alkaloids.

Table 3. Variation of the iodide in the Pd-catalysed cyclisation reaction.[a,b]



[a] Conditions: allene **6a** or **6b**, (0.1 mmol); ArI, (0.1 mmol); $Pd(OAc)_2$, (0.01 mmol); XantPhos, (0.01 mmol); K_2CO_3, (0.23 mmol); MeCN, (3 mL); reflux, 18 h. [b] Isolated yields; the ratio established by ^1H NMR.

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Scheme 5. Synthesis of longamide derivative **17**. a. ArI, $Pd(OAc)_2$, XantPhos, K_2CO_3, MeCN, 90 °C, 26 %; b. MnO_2, THF, 85°C, 56 %; c. (i) 9-BBN, THF, then NaOH, H_2O_2, (ii) anisole, TFA, 60°C, 32 %; d. NBS, THF, r.t., 75 %; e. NaOH, THF/H_2O, r.t., 99 %; f. TEMPO, $PhI(OAc)_2$, $NaHCO_3$, MeCN/H_2O, r.t., 83 %.

Although both metal catalysed processes are relatively efficient, the significant difference in diastereoselectivity of the cyclisation steps was quite intriguing. Introduction of an additional substituent by Pd-catalysed processes did not create opportunity to improve the stereoselectivity due to additional steric requirements, but it rather had the opposite effect. Further to this, it seemed that variation of the aryl substituent did not influence selectivity at all. Inspection of the molecular models suggested likely transition states for the cyclisation step and they are delineated in Figure 1 and Figure 2 The stereoselectivity of Ag-catalysed process (Table 1) was significantly better than that of the corresponding Pd-catalysed transformation (Table 3). Mechanistically these two processes differ and therefore involve unlike intermediates. While Pd-catalysis created π-allylpalladium functionality which then underwent the cyclisation, Ag-promoted transformation resided on coordination of the allene by Ag-salt prior to the cyclisation. As expected and as experimental results demonstrated, the product of the favoured 6-exo cyclisation was the only one obtained. In the case of Pd-catalysed cyclisation (Figure 1) two most likely modes of cyclisation involve either *Re* face (Figure 1a) or *Si* face (Figure 1b) of the prochiral, bond forming C-atom of the π-allylpalladium moiety. Although transition state resulting from the nucleophilic attack from the *Si* face (Figure 1b) would be expected to be less favoured due to 1,3-diaxial interactions of the π-allylpalladium group and the pseudoaxial pyrrolidine C(2)H/C(5)H bonds, what would consequently result in some degree of selectivity, that was not the observed outcome. The stereoselectivity of the cyclisation step is insignificant and two stereoisomers, in all cases, were isolated in almost equal amounts. This may be attributed to the stereochemical properties of the allyl moiety which is planar with the aromatic ring positioned in the same plane to minimize the steric interactions with the pseudoaxial C-H bonds. On the other hand, the diaxial interactions in transition state *b* (Figure 1) may be paralleled by steric interactions between the pseudoequatorial allyl group and the $C(5)H_2$ substituent of the proline ring in transition state *a* (Figure 1). This model might also account for insignificant effect of the aryl group on the diastereoselectivity.

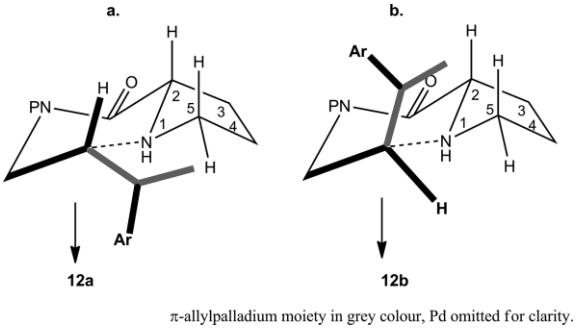


Figure 1. Transition state for the Pd-catalysed cyclisation reaction.

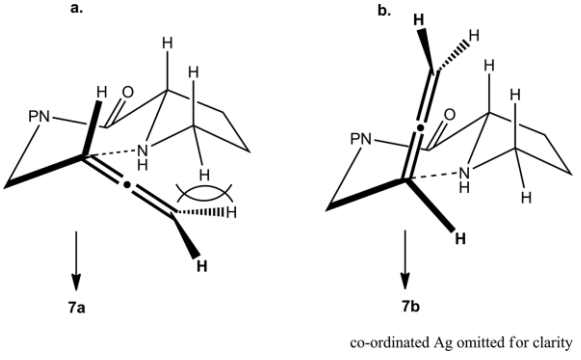


Figure 2. Transition state for the Ag-catalysed cyclization reaction.

Inspection of molecular models of the transition state for the Ag-catalysed transformation revealed possible origin of the observed results (Figure 2). It is likely that detrimental steric interactions are created when the allenyl moiety is orientated pseudoequatorially (Figure 2a). In that position allenic fragment locates the terminal methylene group in close proximity to the proline $C(5)H_2$. These repulsions may outweigh potential 1,3-diaxial interactions of the pseudoaxial allene (Figure 2b) favouring more stable transition state *b* (Figure 2) and formation of *trans* product **7b** as a major component.

Conclusions

In conclusion, our study demonstrated that intramolecular amination of the proline derived allene is a viable route for preparation of the bicyclic derivatives possessing bridgehead nitrogen. Two processes, Pd- or Ag-catalysed cyclisations afforded the expected products in comparable yields although Ag-catalysis showed higher degree of diastereoselectivity. Bicyclic product accessible by the reported methodology can be further transformed and this route was utilized in the synthesis of longamide B, stylisine D and their derivatives.

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Experimental Section

General Information: The NMR spectra were recorded on a Bruker Ascend 400 (400 MHz) spectrometer. Chemical shifts are given in parts per million (δ) downfield from tetramethylsilane as the internal standard. Deuteriochloroform was used as a solvent, unless otherwise stated. Mass spectroscopic data were recorded using an Agilent Technologies 6520 Q-TOF spectrometer coupled with an Agilent 1200 HPLC system or an Agilent Technologies 5975C MS system coupled with an Agilent Technologies 6890N GC system. IR spectra were recorded on an IR Thermo Scientific NICOLET iS10 (4950) spectrometer. Optical rotations were measured on an AUTOPOL IV Automatic Polarimeter Rudolph Research Analytical. $[\alpha]_D$ values are given in deg mL g^{-1} dm^{-1}. Flash chromatography employed silica gel 60 (230–400 mesh) while thin layer chromatography was carried out using alumina plates with a 0.25 mm silica layer (Kieselgel 60 F254, Merck). Compounds were visualized by staining with potassium permanganate solution. The solvents were purified by distillation before use.

***tert*-Butyl 4-Methoxybenzyl(prop-2-yn-1-yl)carbamate (2a):** The title compound was prepared following the procedures described in the literature starting from *p*-methoxybenzylamine (4.20 g, 30.66 mmol) to afford **2a** in a 79 % yield (6.67 g, 24.25 mmol) as a clear oil.[48,56] Spectral data were in accordance with the reported ones.[55]

***tert*-Butyl Benzyl(prop-2-yn-1-yl)carbamate (2b):** The title compound was prepared following the procedure for **2a** starting from benzylamine (3.00 g, 28.04 mmol) to afford **2b** in a 72 % yield (4.95 g, 20.19 mmol) as a clear oil. Spectral data were in accordance with the reported ones.[48]

***tert*-Butyl Buta-2,3-dien-1-yl(4-methoxybenzyl)carbamate (3a):** The title compound was prepared following the procedures described in the literature starting from **2a** (5.00 g, 18.18 mmol) to afford **3a** in a 72 % yield (3.79 g, 13.11 mmol) as a pale yellow oil. Spectral data were in accordance with the reported ones.[57]

***tert*-Butyl Benzyl(buta-2,3-dien-1-yl)carbamate (3b):** The title compound was prepared following the procedure for **3a** starting from **2b** (4.95 g, 20.19 mmol) in a 69 % yield (3.61 g, 13.93 mmol) as a pale yellow oil. Spectral data were in accordance with the reported ones.[57]

***N*-(4-Methoxybenzyl)buta-2,3-dien-1-amine (4a):** To a stirred solution of **3a** (3.41 g, 11.80 mmol) in dichloromethane (25 mL), TFA (9.0 mL, 118 mmol) was added and left stirring for 12 h at room temperature. The reaction mixture was evaporated, redissolved in ethyl acetate (30 mL) and washed with saturated $NaHCO_3$ (3 × 30 mL) and with brine (30 mL). Organic layer was dried with anhydrous Na_2SO_4 and the solvent evaporated under reduced pressure to afford **4a** in a 96 % yield (2.14 g, 11.32 mmol) as light brown oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.24 (d, J = 8.4 Hz, 2H), 6.86 (d, J = 8.5 Hz, 2H), 5.23 (p, J = 6.5 Hz, 1H), 4.83–4.73 (m, 2H), 3.80 (s, 3H), 3.76 (s, 2H), 3.26 (dt, J = 6.0, 2.9 Hz, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 208.46, 158.69, 132.22, 129.45, 113.80, 89.23, 76.02, 67.10, 55.27, 52.44, 47.04. IR (ATR): $\tilde{\nu}$ = 2997, 2908, 2834, 1955, 1511, 831 cm^{-1}. HRMS (ESI/Q-TOF) m/z calcd. for $[C_{12}H_{15}NO + H^+]$: 190.1226, found 190.1230.

***N*-Benzylbuta-2,3-dien-1-amine (4b):** The title compound was prepared following the procedure for **4a** starting from **3b** (923 mg, 3.56 mmol) and TFA (2.73 mL, 35.6 mmol) in a 99 % yield (561 mg, 3.53 mmol) as brown oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.42–7.30(m, 5H), 5.22 (p, J = 6.5 Hz, 1H), 4.87–4.74 (m, 2H), 3.82 (s, 2H), 3.27 (dt, J = 6.2, 3.0 Hz, 2H), 2.65 (s, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 208.61, 139.34, 128.45, 128.38, 127.16, 88.66, 76.15, 52.78, 46.97. IR (ATR): $\tilde{\nu}$ = 3026, 2835, 1954, 1674, 1453, 1120 cm^{-1}. HRMS (ESI/Q-TOF) m/z calcd. for $[C_{11}H_{13}N + H^+]$: 160.1121, found 160.1125.

(S)-*tert*-Butyl 2-(Buta-2,3-dien-1-yl(4-methoxybenzyl)carbamoyl)pyrrolidine-1-carboxylate (5a): To a stirred solution of boc-L-proline (2.13 g, 9.89 mmol) in anhydrous dichloromethane (30 mL), under nitrogen atmosphere, EDCI (2.16 g, 1.35 mmol), and HOBt (1.80 g, 1.35 mmol) and DIPEA (3.12 mL, 19.78 mmol) were added. The mixture was cooled to 0°C and **4a** (1.70 g, 9.00 mmol) was added. Solution was stirred for 12 h at room temperature. The reaction mixture was concentrated under reduced pressure and purified by chromatography on silica gel to afford **5a** in an 88 % yield (3.36 g, 8.70 mmol) as a yellow oil. TLC: R_f 0.50 (PE/EtOAc = 1:1). ^{1}H NMR (400 MHz, $CDCl_3$) mixture of rotamers δ = 7.33–7.15 (m, 2H), 6.93–6.80 (m, 2H), 5.23–5.04 (m, 1H), 4.91–4.82 (m, 1H), 4.79–4.73 (m, 1H), 4.73–4.62 (m, 1H), 4.60–4.53 (m, 1H), 4.52–4.42 (m, 0.5H), 4.30 (d, J = 14.4 Hz, 0.5H), 4.03–3.91 (m, 1H), 3.86–3.74 (m, 3H), 3.72–3.57 (m, 1H), 3.55–3.37 (m, 1H), 2.25–2.06 (m, 2H), 2.01–1.76 (m, 3H), 1.51–1.45 (m, 6H), 1.33 (s, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) mixture of rotamers δ = 209.60, 209.29, 208.87, 208.65, 173.45, 173.13, 173.04, 172.78, 159.25, 159.09, 158.82, 154.52, 153.93, 153.78, 130.18, 129.60, 129.53, 129.36, 128.81, 128.58, 128.33, 114.20, 113.93, 113.90, 87.46, 86.92, 86.43, 86.13, 79.81, 79.68, 79.41, 56.28, 56.02, 55.29, 55.25, 48.40, 47.00, 46.96, 46.88, 45.30, 45.24, 31.44, 31.38, 30.30, 28.56, 28.51, 28.34, 24.47, 23.54. IR (ATR): $\tilde{\nu}$ = 2974, 1954, 1693, 1394, 1162, 847 cm^{-1}. HRMS (ESI/Q-TOF) m/z calcd. for $[C_{22}H_{30}N_2O_4 + H^+]$: 387.2278, found 387.2285.

(S)-*tert*-Butyl 2-(Benzyl(buta-2,3-dien-1-yl)carbamoyl)pyrrolidine-1-carboxylate (5b): The title compound was prepared following the procedure for **5a** starting from **4b** (470 mg, 2.96 mmol), Boc-L-Proline (700 mg, 3.25 mmol), EDCI (682 mg, 3.55 mmol), HOBt (480 mg, 3.55 mmol) and DIPEA (1.03 mL, 5.92 mmol) in a 76 % yield (800 mg, 2.25 mmol) as a pale yellow oil. TLC: R_f 0.30 (PE/EtOAc = 7:3). ^{1}H NMR/^{13}C NMR – attempts were not made to assign the signals as both spectra (enclosed in SI) suggested formation of complex mixture of rotamers; full characterization was carried out for the product **(6b)** obtained in the following step. IR (ATR): $\tilde{\nu}$ = 2975, 2876, 1954, 1694, 1396, 1163 cm^{-1}. $[\alpha]_D$ = +29.0 (c 0.10, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{21}H_{28}N_2O_3 + Na^+]$: 379.1992, found 379.2001.

(S)-*N*-(Buta-2,3-dien-1-yl)-*N*-(4-methoxybenzyl)pyrrolidine-2-carboxamide (6a): To a stirred solution of **5a** (3.00 g, 7.77 mmol) in dichloromethane (20 mL), TFA (5.90 mL, 77.7 mmol) was added and left stirring for 12 h at room temperature. The reaction mixture was evaporated, redissolved in ethyl acetate (30 mL) and washed with saturated $NaHCO_3$ (3 × 30 mL) and with brine (1 × 30 mL). Organic layer was dried with anhydrous Na_2SO_4 and the solvent evaporated under reduced pressure to afford **6a** in a 99 % yield (2.20 g, 7.69 mmol) as a brown oil. ^{1}H NMR (400 MHz, $CDCl_3$) mixture of rotamers δ = 7.19–7.08 (m, 2H), 6.92–6.82 (m, 2H), 5.41 (bs, 1.5H), 5.15–5.02 (m, 1H), 4.90 (s, 1H), 4.81–4.77 (m, 0.5H), 4.68 (d, J = 14.5 Hz, 0.5H), 4.58 (d, J = 16.2 Hz, 0.5H), 4.49–4.38 (m, 1H), 4.27–4.20 (m, 1H), 3.93–3.70 (m, 5H), 3.33–3.23 (m, 1H), 3.13–2.96 (m, 1H), 2.31–2.08 (m, 1H), 1.97–1.73 (m, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) mixture of rotamers δ = 209.05, 208.79, 172.76, 172.56, 159.33, 159.13, 129.65, 128.72, 128.25, 127.49, 114.36, 114.08, 86.30, 77.99, 58.28, 58.12, 55.32, 55.27, 49.25, 47.98, 47.34, 47.29, 44.37, 43.63, 31.16, 31.01, 26.16, 26.11. IR (ATR): $\tilde{\nu}$ = 2956, 1955, 1645, 1512, 1247, 848 cm^{-1}. $[\alpha]_D$ = –57.0 (c 0.10, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{17}H_{22}N_2O_2 + H^+]$: 287.1754, found 287.1762.

(S)-*N*-Benzyl-*N*-(buta-2,3-dien-1-yl)pyrrolidine-2-carboxamide (6b): The title compound was prepared following the procedure for

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6a starting from **5b** (400 mg, 1.12 mmol) and TFA (0.86 mL, 11.2 mmol) in 92 % yield (264 mg, 1.03 mmol) as a dark orange oil. ^{1}H NMR (400 MHz, CDCl3) mixture of rotamers δ = 7.40–7.27 (m, 3H), 7.25–7.18 (m, 2H), 5.17–5.04 (m, 1H), 4.93–4.85 (m, 1H), 4.81–4.73 (m, 1.5H), 4.65 (d, *J* = 16.7 Hz, 0.5H), 4.50 (d, *J* = 15.4 Hz, 1H), 4.23–4.14 (m, 0.5H), 4.13–4.07 (m, 0.5H), 4.06–3.99 (m, 0.5H), 3.94–3.72 (m, 1.5H), 3.35–3.19 (m, 2H), 3.01–2.85 (m, 1H), 2.24–2.14 (m, 0.5H), 2.11–2.01 (m, 0.5H), 1.94–1.69 (m, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) mixture of rotamers δ = 209.09, 208.80, 173.97, 173.74, 136.95, 136.01, 128.97, 128.68, 128.21, 127.83, 127.55, 126.77, 86.40, 77.91, 58.44, 58.28, 49.70, 48.50, 47.72, 47.67, 44.58, 43.94, 31.38, 31.28, 26.53, 26.48. IR (ATR): $\tilde{\nu}$ = 3300, 2964, 2871, 1964, 1640, 1219 cm^{-1}. $[\alpha]_D$ = –237.0 (c 0.10, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{16}H_{20}N_2O + H^+]$: 257.1648, found 257.1658.

(4R,8aS)-2-Benzyl-4-vinylhexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (7a) and (4S,8aS)-2-Benzyl-4-vinylhexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (7b): 6b (50 mg, 0.195 mmol), was dissolved in a mixture of acetone and water (5 mL, 1:1) in a foil-wrapped round-bottom flask. Calcium-carbonate (15.6 mg, 0.156 mmol) and silver-nitrate (26.6 mg, 0.156 mmol) were added, and the mixture was stirred overnight at room temperature. The mixture was thereupon filtered and acetone was removed under reduced pressure. The reaction mixture should not at any point be heated over 30 °C due to the formation of a purple colored side product. The remaining water solution was extracted with diethyl ether (3 × 5 mL), dried with anhydrous Na_2SO_4, concentrated under reduced pressure and purified by chromatography on silica gel to afford **7a** in a 14 % yield (7 mg, 0.027 mmol) as a pale yellow oil and **7b** in a 69 % yield (34.4 mg, 0.134 mmol) as a pale yellow oil which quickly turns purple upon heat and air exposure. TLC: R_{f7a} 0.55 (EtOAc/MeOH = 4:1) and R_{f7b} 0.40 (EtOAc/MeOH = 4:1).

7a: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.35–7.26 (m, 4H), 7.25–7.23 (m, 1H), 5.72 (ddd, *J* = 17.4, 10.2, 7.4 Hz, 1H), 5.28–5.12 (m, 2H), 4.74 (d, *J* = 14.6 Hz, 1H), 4.41 (d, *J* = 14.6 Hz, 1H), 3.26–3.15 (m, 2H), 3.12–3.07 (m, 1H), 3.03–2.96 (m, 2H), 2.28–2.17 (m, 2H), 2.05–1.93 (m, 1H), 1.88–1.73 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 169.86, 136.91, 136.33, 128.62, 128.18, 127.42, 118.62, 65.18, 62.05, 50.89, 50.23, 49.31, 26.46, 21.42. IR (ATR): $\tilde{\nu}$ = 3465, 2919, 1639, 1494, 926, 700 cm^{-1}. $[\alpha]_D$ = –138.0 (c 0.05, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{16}H_{20}N_2O + H^+]$: 257.1648, found 257.1660.

7b: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.36–7.26 (m, 4H), 7.25–7.23 (m, 1H), 5.72–5.57 (m, 1H), 5.23–5.14 (m, 2H), 4.61 (q, *J* = 14.6 Hz, 2H), 3.78 (t, *J* = 8.5 Hz, 1H), 3.25–3.12 (m, 3H), 2.96–2.89 (m, 1H), 2.85–2.78 (m, 1H), 2.34–2.24 (m, 1H), 2.08–1.99 (m, 1H), 1.89–1.69 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 170.88, 136.83, 136.77, 128.67, 128.17, 127.56, 118.37, 61.50, 59.21, 52.52, 49.90, 49.68, 28.70, 22.86. IR (ATR): $\tilde{\nu}$ = 3466, 2917, 1640, 1494, 927, 700 cm^{-1}. $[\alpha]_D$ = –29.0 (c 0.10, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{16}H_{20}N_2O + H^+]$: 257.1648, found 257.1658.

(4R,8aS)-2-(4-Methoxybenzyl)-4-vinylhexahydropyrrolo[1,2-a]-pyrazin-1(2H)-one (7d) and (4S,8aS)-2-(4-Methoxybenzyl)-4-vinylhexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (7c): The title compounds were prepared following the procedure for **7a**/**7b** starting from **6a** (1.3 g, 4.54 mmol), $CaCO_3$ (363 mg, 3.63 mmol) and $AgNO_3$ (618 mg, 3.63 mmol) to afford **7d** in a 14 % yield (184 mg, 0.64 mmol) as a pale yellow oil and **7c** in a 69 % yield (896 mg, 3.13 mmol) as a pale yellow oil which quickly turns purple upon heat and air exposure. TLC: R_{f7d} 0.35 (EtOAc/MeOH = 9:1), R_{f7c} 0.20 (EtOAc/MeOH = 9:1).

7d:* ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.19 (d, *J* = 8.5 Hz, 2H), 6.85 (d, *J* = 8.6 Hz, 2H), 5.71 (ddd, *J* = 10.2, 8.8, 5.4 Hz, 1H), 5.25 (d, *J* = 17.2 Hz, 1H), 5.17 (d, *J* = 10.3 Hz, 1H), 4.67 (d, *J* = 14.4 Hz, 1H), 4.34 (d, *J* = 14.4 Hz, 1H), 3.79 (s, 3H), 3.25–3.12 (m, 2H), 3.11–3.05 (m, 1H), 3.02–2.93 (m, 2H), 2.22 (dt, *J* = 16.8, 8.2 Hz, 2H), 1.98 (ddd, *J* = 21.7, 11.4, 8.4 Hz, 1H), 1.88–1.73 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 169.75, 159.07, 136.30, 129.63, 128.99, 118.80, 114.05, 65.19, 62.12, 55.28, 50.66, 50.30, 48.68, 26.43, 21.40.

7c: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.19 (d, *J* = 8.5 Hz, 2H), 6.85 (d, *J* = 8.6 Hz, 2H), 5.64 (ddd, *J* = 17.0, 10.2, 6.6 Hz, 1H), 5.23–5.14 (m, 2H), 4.54 (q, *J* = 14.4 Hz, 2H), 3.80 (s, 3H), 3.76 (t, *J* = 8.6 Hz, 1H), 3.25–3.10 (m, 3H), 2.97–2.87 (m, 1H), 2.85–2.77 (m, 1H), 2.33–2.22 (m, 1H), 2.07–1.96 (m, 1H), 1.87–1.68 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 170.71, 159.13, 136.79, 129.55, 128.84, 118.43, 114.06, 61.50, 59.20, 55.28, 52.49, 49.61, 49.04, 28.68, 22.82. IR (ATR): $\tilde{\nu}$ = 2956, 2835, 1644, 1512, 1246, 1033 cm^{-1}. $[\alpha]_D$ = –25.0 (c 0.10, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{17}H_{22}N_2O_2 + H^+]$: 287.1754, found 287.1764.

*Due to its low stability the compound was oxidized as described below (compound **8**) and stored and further characterized as other enantiomer of **8** named **8a**.

(S)-2-(4-Methoxybenzyl)-4-vinyl-3,4-dihydropyrrolo[1,2-a]pyrazin-1(2H)-one (8): To a flame-dried round-bottom flask was added **7c** (800 mg, 2.80 mmol), freshly distilled THF (20 mL) and MnO_2 (6.16 g, 70.80 mmol) were added. The reaction was heated to 85 °C and stirred overnight. The reaction mixture was filtered through a pad of Celite. The Celite was washed with hot THF and then placed again into a round-bottom flask and refluxed in THF for 30 minutes. The mixture was then filtered through a new pad of Celite and the combined filtrate concentrated under reduced pressure and purified by chromatography on silica gel to afford **8** in a 57 % yield (450 mg, 1.60 mmol) as a clear pale yellow oil. TLC: R_f 0.28 (Et_2O/PE = 3:1). ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.24 (d, *J* = 8.5 Hz, 2H), 7.01 (dd, *J* = 3.7, 1.4 Hz, 1H), 6.85 (d, *J* = 8.6 Hz, 2H), 6.71 (s, 1H), 6.32–6.21 (m, 1H), 5.77 (ddd, *J* = 17.1, 10.2, 7.0 Hz, 1H), 5.33 (d, *J* = 10.2 Hz, 1H), 5.12 (d, *J* = 17.0 Hz, 1H), 4.77–4.58 (m, 3H), 3.79 (s, 3H), 3.53 (dd, *J* = 12.7, 4.5 Hz, 1H), 3.40 (dd, *J* = 12.7, 7.6 Hz, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 159.33, 159.12, 133.70, 129.72, 129.04, 124.41, 121.95, 120.71, 114.02, 109.98, 56.76, 55.28, 49.88, 48.55. IR (ATR): $\tilde{\nu}$ = 2911, 2836, 1634, 1511, 1242, 733 cm^{-1}. $[\alpha]_D$ = –1.3 (c 0.30, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{17}H_{18}N_2O_2+ H^+]$: 283.1441, found 283.1449.

8a: IR (ATR): $\tilde{\nu}$ = 2911, 2836, 1634, 1512, 1243, 732 cm^{-1}. $[\alpha]_D$ = +1.0 (c 0.20, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{17}H_{18}N_2O_2 + H^+]$: 283.1441, found 283.1455.

(S)-4-(2-Hydroxyethyl)-2-(4-methoxybenzyl)-3,4-dihydropyrrolo[1,2-a]pyrazin-1(2H)-one (9a) and (S)-2-(1-Oxo-1,2,3,4-tetrahydropyrrolo[1,2-a]pyrazin-4-yl)ethyl 2,2,2-Trifluoroacetate (9): To a solution of **8** (400 mg, 1.42 mmol) in THF (15 mL), a solution of 9-BBN in THF (0.5 M, 4.26 mmol, 8.51 mL) was added. The reaction was monitored by TLC for the consumption of starting alkene and was completed after 3 hours. Thereafter the reaction mixture was cooled to 0 °C, and NaOH (1.0 M, 12.76 mL) and H_2O_2 (35 wt.-%/vol, 1.54 mL) were added. The reaction was thereafter stirred overnight at room temperature. The reaction was then transferred into a separatory funnel with saturated $NaHCO_3$ (10 mL) and extracted with ethyl acetate (3 × 10 mL). The combined organic fractions were dried, filtered, concentrated under reduced pressure and purified by chromatography on silica gel to afford 352 mg of clear colorless oil, slightly contaminated with 9-BBN residues (**9a**). TLC: R_f 0.30 (EtOAc/benzene = 1:1). The oil was used without further purification in the following step. Under nitrogen atmosphere the crude oil was dissolved in TFA (5 mL) and anisole (0.26 mL, 2.37 mmol) was

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added. The reaction was stirred at 60 °C overnight. The reaction mixture was evaporated and the remaining residue diluted with ethyl acetate (15 mL) and transferred into a separatory funnel and washed with $NaHCO_3$ (3 × 15 mL). The combined organic layer was dried, filtered, concentrated under reduced pressure and purified by chromatography on silica gel to afford **9** in a 55 % yield over 2 steps (216 mg, 0.78 mmol) as a pale yellow oil. TLC: R_f 0.35 (EtOAc = 100 %).

9a: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.24 (d, J = 8.4 Hz, 2H), 6.95 (d, J = 3.2 Hz, 1H), 6.85 (d, J = 8.4 Hz, 2H), 6.80 (s, 1H), 6.23–6.14 (m, 1H), 4.90 (d, J = 14.4 Hz, 1H), 4.42–4.29 (m, 2H), 3.83–3.74 (m, 4H), 3.54–3.46 (m, 1H), 3.45–3.38 (m, 1H), 3.30 (dd, J = 12.8, 2.6 Hz, 1H), 1.88–1.83 (m, 1H), 1.74–1.65 (m, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 159.39, 159.20, 129.97, 129.12, 128.34, 123.69, 122.59, 114.07, 113.87, 109.66, 58.61, 55.31, 50.85, 49.00, 48.56, 35.58. HRMS (ESI/Q-TOF) m/z calcd. for [$C_{17}H_{20}N_2O_3$ + H^+]: 301.1547, found 301.1555.

9: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 6.99 (d, J = 3.0 Hz, 1H), 6.85–6.72 (m, 2H), 6.31–6.24 (m, 1H), 4.55–4.45 (m, 1H), 4.40–4.32 (m, 1H), 4.17–4.09 (m, 1H), 3.98 (dd, J = 12.8, 3.9 Hz, 1H), 3.50–3.42 (m, 1H), 2.42–2.30 (m, 1H), 2.22 (ddd, J = 14.9, 9.9, 5.2 Hz, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 161.15, 157.16, 134.77, 123.26, 114.67, 110.25, 64.44, 50.82, 44.76, 31.63. IR (ATR): $\tilde{\nu}$ = 2884, 1784, 1634, 1552, 1334, 741 cm^{-1}. $[\alpha]_D$ = –57.6 (c 0.34, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{11}H_{11}F_3N_2O_3$ + H^+]: 277.0795, found 277.0792.

(S)-2-(6,7-Dibromo-1-oxo-1,2,3,4-tetrahydropyrrolo[1,2-a]pyrazin-4-yl)ethyl 2,2,2-Trifluoroacetate (10): To a solution of **9** (120 mg, 0.44 mmol) in THF (5 mL), was added a solution of NBS (162 mg, 0.913 mmol) in THF (1 mL) at room temperature. The reaction was stirred overnight at room temperature. The reaction mixture was evaporated under reduced pressure and the residue purified by chromatography on silica gel to afford **10** in an 85 % yield (160 mg, 0.37 mmol) as a pale-yellow oil. TLC: R_f 0.40 (EtOAc = 100 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.02 (s, 1H), 4.52–4.46 (m, 1H), 4.45–4.40 (m, 2H), 3.96 (dd, J = 13.2, 3.9 Hz, 1H), 3.56 (dd, J = 12.9, 4.7 Hz, 1H), 2.40–2.26 (m, 1H), 2.26–2.13 (m, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 159.41, 157.21, 124.74, 116.56, 106.94, 101.42, 64.16, 51.05, 43.32, 31.04. IR (ATR): $\tilde{\nu}$ = 2927, 1651, 1428, 1337, 943, 750 cm^{-1}. $[\alpha]_D$ = –18.6 (c 0.50, CH_3OH). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{11}H_9Br_2F_3N_2O_3$ + H^+]: 432.9005, found 432.9001.

(S)-6,7-Dibromo-4-(2-hydroxyethyl)-3,4-dihydropyrrolo[1,2-a]pyrazin-1(2H)-one (11): To a stirred solution of **10** (50 mg, 0.115 mmol) in a mixture of methanol, THF and water (5 mL, 1:2:1), NaOH (0.46 mL of 1 M solution) was added dropwise. After 3 h methanol and THF were evaporated under reduced pressure and the remaining water solution was extracted with ethyl acetate (3 × 5 mL). The combined organic layer was dried, filtered, concentrated under reduced pressure to afford **11** in a 99 % yield (38.5 mg, 0.114 mmol) as a pale-yellow oil. TLC: R_f 0.40 (EtOAc/MeOH = 9:1). ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.01 (s, 1H), 5.91 (s, 1H), 4.61–4.47 (m, 1H), 3.90 (dd, J = 12.9, 3.9 Hz, 1H), 3.78 (s, 2H), 3.69 (dd, J = 12.6, 5.0 Hz, 1H), 2.16–2.05 (m, 1H), 1.97–1.87 (m, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 159.03, 121.81, 116.13, 106.66, 100.84, 59.26, 52.16, 43.31, 34.47. IR (ATR): $\tilde{\nu}$ = 3306, 2360, 1645, 1549, 1338, 1063 cm^{-1}. $[\alpha]_D$ = –27.0 (c 0.10, CH_3OH). HRMS (ESI/Q-TOF) m/z calcd. for [$C_9H_{10}Br_2N_2O_2$ + H^+]: 336.9182, found 336.9188.

Longamide B: To a stirred solution of **11** (15 mg, 0.045 mmol) in a mixture of acetonitrile and water (3 mL, 1:1), iodobenzene diacetate (BAIB) (36 mg, 0.112 mmol), $NaHCO_3$ (15 mg, 0.176 mmol) and 2,2,6,6-tetramethyl-piperidine-1-oxyl (TEMPO) (1.6 mg, 0.011 mmol) were added. The mixture was stirred for 18 h at room temperature. Upon completion saturated solution of $NaHCO_3$ (5 mL) was added to the reaction mixture and the aqueous phase was washed with ethyl acetate (3 × 5 mL). The remaining water solution was acidified with 2 M HCl to pH 2 and extracted with ethyl acetate (3 × 5 mL). The combined organic layer was dried with anhydrous Na_2SO_4, filtered and concentrated under reduced pressure. The crude white solid was washed with the mixture of dichloro methane and petroleum ether (2:8), and recrystalysed from the mixture of dichloromethane and methanol (98:2) to afford **Longamide B** in an 83 % yield (12.5 mg, 0.037 mmol) as a white crystalline solid. ^{1}H NMR (400 MHz, 98 % $CDCl_3$ + 2 % MeOD) δ = 7.00 (s, 1H), 4.75 (d, J = 11.6 Hz, 1H), 3.89 (d, J = 13.5 Hz, 1H), 3.70 (d, J = 13.5 Hz, 1H), 2.97–2.89 (m, 1H), 2.56 (d, J = 16.6 Hz, 1H). ^{13}C NMR (101 MHz, 98 % $CDCl_3$ + 2 % MeOD) δ = 171.90, 159.57, 124.68, 116.39, 106.88, 101.29, 50.53, 42.86, 35.38. IR (ATR): $\tilde{\nu}$ = 3305, 2359, 1634, 1338, 1059, 749 cm^{-1}. $[\alpha]_D$ = –5.5 (c 0.20, CH_3OH). HRMS (ESI/Q-TOF) m/z calcd. for [$C_9H_8Br_2N_2O_3$ + H^+]: 350.8974, found 350.8971.

Stylisine D: To a stirred solution of longamide B (10 mg, 0.028 mmol) in THF (5 mL), EDCI (8.2 mg, 0.043 mmol), HOBt (5.8 mg, 0.043 mmol) and DIPEA (8.4 μL, 0.048 mmol) were added at 0 °C. After stirring for 5 minutes, the solution of 25 % ammonium hydroxide (6.8 mg, 0.048 mmol) was added and the solution was stirred for 18 h at room temperature. Upon completion the solvents were evaporated under reduced pressure, residues redissolved in ethyl acetate (5 mL) and washed with HCl 0.5 M (3 × 5 mL), then with $NaHCO_3$ (3 × 5 mL) and finally with brine (2 × 5 mL). The organic layer was dried with anhydrous Na_2SO_4 concentrated under reduced pressure and the residue purified by chromatography on silica gel to afford stylisine D in an 81 % yield (8.1 mg, 0.023 mmol) as a pale yellow amorphous solid. TLC: R_f 0.15 (EtOAc/MeOH = 9:1). ^{1}H NMR (400 MHz, 98 % $CDCl_3$ + 2 % MeOD) δ = 6.99 (s, 1H), 4.82 (d, J = 10.6 Hz, 1H), 3.87 (d, J = 12.7 Hz, 1H), 3.63 (d, J = 13.2 Hz, 1H), 2.84–2.73 (m, 1H), 2.49 (d, J = 14.9 Hz, 1H). ^{13}C NMR (101 MHz, 98 % $CDCl_3$ + 2 % MeOD) δ = 171.69, 124.72, 116.43, 107.10, 101.23, 50.96, 42.94, 36.41. IR (ATR): $\tilde{\nu}$ = 3381, 1635, 1349, 1331, 999, cm^{-1}. $[\alpha]_D$ = –7.0 (c 0.10, CH_3OH). HRMS (ESI/Q-TOF) m/z calcd. for [$C_9H_9Br_2N_3O_2$ + H^+]: 349.9134, found 349.9139.

(4R,8aS)-4-(1-(3-Chloro-4-methylphenyl)vinyl)-2-(4-methoxybenzyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12aa) and (4S,8aS)-4-(1-(3-Chloro-4-methylphenyl)vinyl)-2-(4-methoxybenzyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12ba): To a solution of **6a** (1114 mg, 3.90 mmol) in acetonitrile (10 mL), under the atmosphere of nitrogen, K_2CO_3 (1075 mg, 7.79 mmol), $Pd(OAc)_2$ (87.3 mg, 0.39 mmol), XantPhos (225.5 mg, 0.39 mmol) and 2-chloro-4-iodo-1-methylbenzene (982.8 mg, 3.90 mmol) were added. The solution was refluxed for 18 h. Upon completion the reaction mixture was concentrated under reduced pressure and the residue purified by chromatography on silica gel to afford **12aa** in a 32 % yield (512.9 mg, 1.25 mmol) as a pale-yellow oil and **12ba** in a 26 % yield (416.8 mg, 1.01 mmol) as a pale yellow oil. TLC: R_{f12aa} 0.50 (Et_2O = 100 %) and R_{f12ba} 0.40 (Et_2O = 100 %).

12aa: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.46 (d, J = 1.4 Hz, 1H), 7.32–7.27 (m, 1H), 7.18–7.12 (m, 3H), 6.84 (d, J = 8.6 Hz, 2H), 5.42 (s, 1H), 5.26 (s, 1H), 4.75 (d, J = 14.5 Hz, 1H), 4.21 (d, J = 14.5 Hz, 1H), 3.79 (s, 3H), 3.64 (dd, J = 10.6, 4.6 Hz, 1H), 3.37 (t, J = 11.3 Hz, 1H), 3.19–3.08 (m, 2H), 2.98 (td, J = 8.6, 3.9 Hz, 1H), 2.36 (s, 3H), 2.29–2.18 (m, 2H), 2.03 (dt, J = 12.7, 9.6 Hz, 1H), 1.87–1.71 (m, 2H), 1.64 (s, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 158.99, 145.09, 138.83, 135.62, 134.32, 130.79, 129.44, 128.97, 127.35, 125.03, 117.20, 114.01, 65.26, 62.86, 55.26, 49.84, 49.77, 48.70, 26.86, 21.64, 19.74. IR (ATR): $\tilde{\nu}$ = 2923, 1645, 1512, 1246, 821, 734 cm^{-1}. $[\alpha]_D$ = –55.0 (c 0.10, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{20}H_{29}NO_4$ + H^+]: 411.1834, found 411.1837.

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12ba: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.28 (d, *J* = 1.5 Hz, 1H), 7.14 (d, *J* = 8.6 Hz, 2H), 7.08 (d, *J* = 7.9 Hz, 1H), 7.00 (dd, *J* = 7.9, 1.6 Hz, 1H), 6.86–6.79 (m, 2H), 5.38 (d, *J* = 15.7 Hz, 2H), 4.65 (d, *J* = 14.4 Hz, 1H), 4.31 (d, *J* = 14.3 Hz, 1H), 3.87 (t, *J* = 8.1 Hz, 1H), 3.79 (s, 3H), 3.50 (dd, *J* = 8.1, 4.4 Hz, 1H), 3.22–3.11 (m, 2H), 3.11–3.04 (m, 1H), 2.67–2.57 (m, 1H), 2.34 (s, 3H), 2.28–2.14 (m, 2H), 1.90–1.75 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 171.62, 159.09, 145.47, 139.30, 135.32, 134.27, 130.74, 129.62, 129.55, 128.90, 128.31, 127.13, 126.51, 124.77, 116.26, 114.04, 62.32, 60.98, 55.23, 53.47, 49.32, 49.03, 28.45, 24.00, 19.72. IR (ATR): $\tilde{\nu}$ = 2915, 1646, 1511, 1245, 818, 734 cm^{-1}. $[\alpha]_D$ = –7.0 (c 0.20, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{24}H_{27}ClN_2O_2 + H^+]$: 411.1834, found 411.1836.

(4R,8aS)-4-(1-(3,4-Dichlorophenyl)vinyl)-2-(4-methoxybenzyl)-hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12ab) and (4S,8aS)-4-(1-(3,4-Dichlorophenyl)vinyl)-2-(4-methoxybenzyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12bb): The title compounds were prepared following the procedure for **12aa**/**12ba** starting from **6a** (30.0 mg, 0.105 mmol), K_2CO_3 (29.0, 0.210 mmol), $Pd(OAc)_2$ (2.4 mg. 0.0105 mmol), XantPhos (6.1 mg, 0.0105 mmol) and 1,2-dichloro-4-iodobenzene (28.7 mg, 0.105 mmol) to afford **12ab** in a 33 % yield (15.1 mg, 0.035 mmol) as a yellow oil and **12bb** in a 26 % yield (11.8 mg, 0.028 mmol) as a yellow oil. TLC: R_{f12ab} 0.40 (Et_2O = 100 %), R_{f12bb}0.30 (Et_2O = 100 %).

12ab: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.59 (s, 1H), 7.36 (s, 2H), 7.15 (d, *J* = 8.5 Hz, 2H), 6.84 (d, *J* = 8.5 Hz, 2H), 5.44 (s, 1H), 5.30 (s, 1H), 4.72 (d, *J* = 14.5 Hz, 1H), 4.25 (d, *J* = 14.5 Hz, 1H), 3.79 (s, 3H), 3.65 (dd, *J* = 10.6, 4.7 Hz, 1H), 3.35 (t, *J* = 11.3 Hz, 1H), 3.21–3.07 (m, 2H), 2.92 (td, *J* = 8.4, 4.1 Hz, 1H), 2.30–2.18 (m, 2H), 2.04 (dt, *J* = 12.9, 9.4 Hz, 1H), 1.87–1.71 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 170.23, 159.09, 144.41, 139.46, 132.46, 131.96, 130.26, 129.45, 128.90, 128.84, 126.25, 118.23, 114.07, 65.07, 62.65, 55.28, 49.60, 49.18, 48.76, 26.96, 21.75. IR (ATR): $\tilde{\nu}$ = 2920, 2645, 1511, 1244, 1029, 817 cm^{-1}. $[\alpha]_D$ = –65.7 (c 0.30, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{23}H_{24}Cl_2N_2O_2 + H^+]$: 431.1288, found 431.1303.

12bb: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.38 (d, *J* = 1.9 Hz, 1H), 7.29 (d, *J* = 8.3 Hz, 1H), 7.14 (d, *J* = 8.5 Hz, 2H), 7.03 (dd, *J* = 8.3, 2.0 Hz, 1H), 6.83 (d, *J* = 8.6 Hz, 2H), 5.39 (d, *J* = 24.0 Hz, 2H), 4.67 (d, *J* = 14.4 Hz, 1H), 4.30 (d, *J* = 14.4 Hz, 1H), 3.86 (t, *J* = 7.9 Hz, 1H), 3.80 (s, 3H), 3.46 (dd, *J* = 8.2, 4.2 Hz, 1H), 3.21–3.11 (m, 2H), 3.11–3.03 (m, 1H), 2.59 (dd, *J* = 15.9, 9.1 Hz, 1H), 2.30–2.16 (m, 2H), 1.90–1.75 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 171.58, 159.22, 144.85, 140.08, 132.41, 131.65, 130.20, 129.65, 128.86, 128.67, 126.01, 117.36, 114.11, 62.59, 60.79, 55.27, 53.48, 49.04, 48.86, 28.39, 24.10. IR (ATR): ν 2932, 1645, 1511, 1245, 1029, 818 cm^{-1}. $[\alpha]_D$ = –7.6 (c 0.50, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{23}H_{24}Cl_2N_2O_2 + H^+]$: 431.1288, found 431.1302.

(4R,8aS)-4-(1-(3,5-Dimethoxyphenyl)vinyl)-2-(4-methoxybenzyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12ac) and (4S,8aS)-4-(1-(3,5-Dimethoxyphenyl)vinyl)-2-(4-methoxybenzyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12bc): The title compounds were prepared following the procedure for **12aa**/**12ba** starting from **6a** (30.0 mg, 0.105 mmol), K_2CO_3 (29.0, 0.210 mmol), $Pd(OAc)_2$ (2.4 mg. 0.0105 mmol), XantPhos (6.1 mg, 0.0105 mmol) and 1-iodo-3,5-dimethoxybenzene (27.7 mg, 0.105 mmol) to afford **12ac** in a 29 % yield (12.8 mg, 0.030 mmol) as a yellow oil and **12bc** in a 24 % yield (10.6 mg, 0.025 mmol) as a yellow oil. TLC: R_{f12ac} 0.35 (Et_2O = 100 %), R_{f12bc} 0.25 (Et_2O = 100 %).

12ac: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.16 (d, *J* = 8.5 Hz, 2H), 6.83 (d, *J* = 8.6 Hz, 2H), 6.65 (d, *J* = 2.1 Hz, 2H), 6.41 (d, *J* = 2.1 Hz, 1H), 5.45 (s, 1H), 5.29 (s, 1H), 4.75 (d, *J* = 14.5 Hz, 1H), 4.20 (d, *J* = 14.5 Hz, 1H), 3.82–3.75 (m, 9H), 3.62 (dd, *J* = 10.3, 4.3 Hz, 1H), 3.38 (t, *J* = 11.3 Hz, 1H), 3.16 (dd, *J* = 11.9, 4.6 Hz, 1H), 3.12–3.01 (m, 2H), 2.30–2.18 (m, 2H), 2.02 (dt, *J* = 12.3, 10.0 Hz, 1H), 1.87–1.72 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 170.14, 160.63, 158.97, 146.23, 141.85, 129.45, 129.04, 117.32, 113.98, 105.23, 99.55, 65.42, 63.29, 55.31, 55.25, 50.31, 50.02, 48.65, 26.76, 21.57. IR (ATR): $\tilde{\nu}$ = 2954, 2835, 1651, 1589, 1155, 846 cm^{-1}. $[\alpha]_D$ = –66.7 (c 0.34, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{25}H_{30}N_2O_4 + H^+]$: 423.2278, found 423.2296.

12bc: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.13 (d, *J* = 8.6 Hz, 2H), 6.81 (d, *J* = 8.6 Hz, 2H), 6.44 (d, *J* = 2.1 Hz, 2H), 6.38 (t, *J* = 2.1 Hz, 1H), 5.39 (d, *J* = 8.5 Hz, 2H), 4.59 (d, *J* = 14.4 Hz, 1H), 4.35 (d, *J* = 14.5 Hz, 1H), 3.89 (t, *J* = 8.1 Hz, 1H), 3.79 (s, 3H), 3.76 (s, 6H), 3.53 (dd, *J* = 8.3, 4.2 Hz, 1H), 3.24–3.13 (m, 2H), 3.13–3.06 (m, 1H), 2.66 (dd, *J* = 16.3, 8.7 Hz, 1H), 2.31–2.23 (m, 1H), 2.24–2.14 (m, 1H), 1.85 (ddd, *J* = 27.5, 14.5, 8.3 Hz, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 171.60, 160.62, 160.38, 159.04, 146.91, 142.49, 129.46, 128.96, 125.53, 116.17, 113.99, 105.09, 99.34, 62.30, 61.10, 55.29, 55.21, 53.50, 49.60, 49.13, 30.33, 28.59, 23.95. IR (ATR): $\tilde{\nu}$ = 2935, 2837, 1645, 1589, 1154, 734 cm^{-1}. $[\alpha]_D$ = –9.3 (c 0.30, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{25}H_{30}N_2O_4 + H^+]$: 423.2278, found 423.2298.

(4R,8aS)-4-(1-(4-Fluorophenyl)vinyl)-2-(4-methoxybenzyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12ad) and (4S,8aS)-4-(1-(4-Fluorophenyl)vinyl)-2-(4-methoxybenzyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12bd): The title compounds were prepared following the procedure for **12aa**/**12ba** starting from **6a** (30.0 mg, 0.105 mmol), K_2CO_3 (29.0, 0.210 mmol), $Pd(OAc)_2$ (2.4 mg. 0.0105 mmol), XantPhos (6.1 mg, 0.0105 mmol) and 1-fluoro-4-iodobenzene (23.3 mg, 0.105 mmol) to afford **12ad** in a 39 % yield (15.4 mg, 0.041 mmol) as an orange oil and **12bd** in a 29 % yield (11.7 mg, 0.031 mmol) as an orange oil. TLC: R_{f12ad} 0.40 (Et_2O = 100 %), R_{f12bd}0.35 (Et_2O = 100 %).

12ad: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.52–7.42 (m, 2H), 7.15 (d, *J* = 8.4 Hz, 2H), 6.98 (t, *J* = 8.6 Hz, 2H), 6.83 (d, *J* = 8.4 Hz, 2H), 5.39 (s, 1H), 5.25 (s, 1H), 4.74 (d, *J* = 14.4 Hz, 1H), 4.20 (d, *J* = 14.5 Hz, 1H), 3.79 (s, 3H), 3.64 (dd, *J* = 10.9, 4.6 Hz, 1H), 3.36 (t, *J* = 11.2 Hz, 1H), 3.20–3.08 (m, 2H), 2.98 (td, *J* = 8.6, 3.9 Hz, 1H), 2.28–2.16 (m, 2H), 2.10–1.97 (m, 1H), 1.86–1.69 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 170.26, 159.05, 145.39, 129.47, 129.04, 128.62, 128.54, 128.36, 126.83, 116.90, 115.30, 115.09, 114.03, 65.29, 63.21, 55.27, 49.78, 49.74, 48.75, 26.89, 21.68. IR (ATR): $\tilde{\nu}$ = 2958, 2835, 1651, 1509, 1245, 842 cm^{-1}. $[\alpha]_D$ = –72.1 (c 0.38, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{23}H_{25}FN_2O_2 + H^+]$: 381.1973, found 381.1992.

12bd: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.24–7.10 (m, 4H), 6.92 (t, *J* = 8.6 Hz, 2H), 6.83 (d, *J* = 8.5 Hz, 2H), 5.35 (d, *J* = 24.5 Hz, 2H), 4.69 (d, *J* = 14.4 Hz, 1H), 4.26 (d, *J* = 14.4 Hz, 1H), 3.88 (t, *J* = 7.9 Hz, 1H), 3.80 (s, 3H), 3.47 (dd, *J* = 8.3, 4.2 Hz, 1H), 3.21–3.12 (m, 2H), 3.12–3.04 (m, 1H), 2.66–2.56 (m, 1H), 2.28–2.20 (m, 2H), 1.91–1.75 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 171.67, 159.18, 145.76, 129.69, 129.65, 129.03, 128.30, 128.22, 126.55, 115.83, 115.24, 115.03, 114.07, 62.84, 60.99, 55.28, 53.52, 49.18, 49.00, 28.38, 24.08. IR (ATR): $\tilde{\nu}$ = 2903, 1634, 1508, 1245, 840, 816 cm^{-1}. $[\alpha]_D$ = –2.1 (c 0.47, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for $[C_{23}H_{25}FN_2O_2 + H^+]$: 381.1973, found 381.1992.

(4R,8aS)-2-Benzyl-4-(1-(naphthalen-2-yl)vinyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one(12ae) and (4S,8aS)-2-Benzyl-4-(1-(naphthalen-2-yl)vinyl)hexahydropyrrolo[1,2-a]pyrazin-1(2H)-one (12be): The title compounds were prepared following the procedure for **12aa**/**12ba** starting from **6b** (26.9 mg, 0.105 mmol), K_2CO_3 (29.0, 0.210 mmol), $Pd(OAc)_2$ (2.4 mg. 0.0105 mmol), XantPhos (6.1 mg, 0.0105 mmol) and 2-iodonaphthalene (26.7 mg, 0.105 mmol) to afford **12ae** in a 43 % yield (17.3 mg, 0.045 mmol) as an orange oil and **12be** in a 32 % yield (12.7 mg, 0.033 mmol)

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as an orange oil. TLC: R_{f12ae} 0.60 (Et_2O = 100 %), R_{f12be} 0.40 (Et_2O = 100 %).

12ae: ^1H NMR (400 MHz, $CDCl_3$) δ = 7.96 (s, 1H), 7.83–7.75 (m, 3H), 7.60 (d, J = 7.2 Hz, 1H), 7.51–7.41 (m, 2H), 7.32–7.27 (m, 2H), 7.24–7.20 (m, 2H), 5.58 (s, 1H), 5.38 (s, 1H), 4.83 (d, J = 14.7 Hz, 1H), 4.27 (d, J = 14.7 Hz, 1H), 3.82 (dd, J = 10.7, 4.6 Hz, 1H), 3.48 (t, J = 11.3 Hz, 1H), 3.23 (dd, J = 12.0, 4.7 Hz, 1H), 3.18 (t, J = 8.3 Hz, 1H), 3.10 (td, J = 8.5, 3.6 Hz, 1H), 2.34–2.20 (m, 2H), 2.16–2.02 (m, 1H), 1.89–1.72 (m, 2H), 1.61–1.49 (m, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 170.46, 146.33, 137.03, 136.95, 133.27, 132.96, 128.62, 128.29, 128.06, 127.92, 127.54, 127.41, 126.24, 126.13, 125.68, 125.07, 117.41, 65.47, 63.18, 50.58, 49.98, 49.34, 26.86, 21.67. IR (ATR): $\tilde{\nu}$ = 3058, 2961, 1645, 1453, 752, 700 cm^{-1}. $[\alpha]_D$ = –48.0 (c 0.05, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{26}H_{26}N_2O$ + H^+]: 383.2118, found 383.2123.

12be: ^1H NMR (400 MHz, $CDCl_3$) δ = 7.83–7.77 (m, 1H), 7.75–7.70 (m, 2H), 7.66 (s, 1H), 7.50–7.44 (m, 2H), 7.41 (d, J = 8.6 Hz, 1H), 7.30–7.26 (m, 2H), 7.22–7.16 (m, 2H), 5.53 (d, J = 4.4 Hz, 2H), 4.72 (d, J = 14.6 Hz, 1H), 4.37 (d, J = 14.6 Hz, 1H), 3.95 (t, J = 7.9 Hz, 1H), 3.73 (dd, J = 8.7, 4.0 Hz, 1H), 3.31–3.19 (m, 2H), 3.18–3.12 (m, 1H), 2.75–2.67 (m, 1H), 2.32–2.21 (m, 2H), 1.94–1.80 (m, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 171.85, 146.71, 137.53, 136.98, 133.23, 132.84, 128.67, 128.19, 127.94, 127.56, 127.53, 126.20, 126.04, 125.16, 125.02, 116.42, 62.42, 61.18, 53.65, 50.01, 49.74, 28.55, 24.06. IR (ATR): $\tilde{\nu}$ = 3058, 2922, 1652, 1454, 753, 701 cm^{-1}. $[\alpha]_D$ = 0.0 (c 0.10, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{26}H_{26}N_2O$ + H^+]: 383.2118, found 383.2124.

(S)-4-(1-(3-Chloro-4-methylphenyl)vinyl)-2-(4-methoxybenzyl)-3,4-dihydropyrrolo[1,2-a]pyrazin-1(2H)-one(13): The title compound was prepared following the procedure for **8** starting from **12ba** (320 mg, 0.78 mmol) and MnO_2 (1716 mg, 19.5 mmol) to afford **13** in a 56 % yield (178 mg, 0.44 mmol) as a pale yellow oil. TLC: R_f 0.20 (PE/Et_2O = 1:1). ^1H NMR (400 MHz, $CDCl_3$) δ = 7.13- 7.08 (m, 3H), 7.06–7.02 (m, 2H), 6.83–6.75 (m, 4H), 6.39–6.24 (m, 1H), 5.24 (s, 1H), 5.12–5.06 (s, 1H), 4.76 (d, J = 14.5 Hz, 1H), 4.34–4.22 (m, 2H), 3.80 (s, 3H), 3.75 (dd, J = 12.9, 4.7 Hz, 1H), 3.25 (dd, J = 12.8, 2.8 Hz, 1H), 2.35 (s, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 159.13, 144.51, 137.49, 136.08, 134.64, 131.06, 129.87, 128.92, 128.66, 126.98, 126.50, 125.00, 124.61, 122.88, 116.50, 113.92, 113.72, 110.49, 57.15, 55.24, 48.45, 48.17, 19.76. IR (ATR): $\tilde{\nu}$ = 2912, 2835, 1634, 1243, 1072, 733 cm^{-1}. $[\alpha]_D$ = –72.5 (c 0.20, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{24}H_{23}ClN_2O_2$ + H^+]: 407.1521, found 407.1526.

(S)-4-((S)-1-(3-Chloro-4-methylphenyl)-2-hydroxyethyl)-2-(4-methoxybenzyl)-3,4-dihydropyrrolo[1,2-a]pyrazin-1(2H)-one(14a) and (S)-2-(3-Chloro-4-methylphenyl)-2-((S)-1-oxo-1,2,3,4-tetrahydropyrrolo[1,2-a]pyrazin-4-yl)ethyl 2,2,2-Trifluoroacetate (14): The title compound was prepared following the procedure for **9** starting from **13** (170 mg, 0.42 mmol), 9-BBN (0.5 M solution in THF) (2.52 mL, 1.26 mmol), followed by NaOH (1 M, 3.77 mL) and H_2O_2 (35 wt.-%/vol, 0.46 mL) to afford crude **14a** which was then treated with anisole (90.1 mg, 0.84 mmol) in TFA (5 mL) to afford **14** in an overall 32 % yield (53.9 mg, 0.134 mmol) as a pale yellow oil. TLC: R_{f14a} 0.5 (Et_2O = 100 %), R_{f14} 0.3 (Et_2O = 100 %).

14a: ^1H NMR (400 MHz, $CDCl_3$) δ = 7.13 (d, J = 8.4 Hz, 2H), 7.06 (d, J = 7.7 Hz, 1H), 7.01 (s, 2H), 6.97 (d, J = 2.8 Hz, 1H), 6.82 (d, J = 8.4 Hz, 2H), 6.61 (d, J = 7.6 Hz, 1H), 6.28–6.19 (m, 1H), 4.68 (d, J = 14.4 Hz, 1H), 4.53 (dd, J = 10.9, 3.5 Hz, 1H), 4.28 (d, J = 14.4 Hz, 1H), 3.80 (s, 3H), 3.69 (dd, J = 12.9, 4.1 Hz, 1H), 3.50–3.38 (m, 2H), 3.01 (d, J = 12.9 Hz, 1H), 2.95–2.85 (m, 1H), 2.34 (s, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 159.43, 159.17, 138.47, 135.21, 134.70, 131.26, 130.05, 129.07, 128.58, 126.87, 124.37, 123.50, 114.05, 109.56, 63.04, 55.25, 54.51, 48.72, 48.39, 47.73, 19.71. IR (ATR): $\tilde{\nu}$ = 3376, 2919, 2870, 2161, 1616, 751 cm^{-1}. $[\alpha]_D$ = –91.0 (c 0.10, CH_2Cl_2). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{24}H_{25}ClN_2O_3$ + H^+]: 425.1626, found 425.1637.

14: ^1H NMR (400 MHz, $CDCl_3$) δ = 7.31 (s, 1H), 7.26 (d, J = 7.2 Hz, 1H), 7.10 (d, J = 7.7 Hz, 1H), 7.00 (d, J = 3.5 Hz, 1H), 6.91 (d, J = 4.3 Hz, 1H), 6.85 (s, 1H), 6.38–6.24 (m, 1H), 4.58 (dd, J = 10.9, 2.7 Hz, 1H), 4.40 (dd, J = 11.2, 2.4 Hz, 1H), 4.04 (dd, J = 11.3, 4.8 Hz, 1H), 3.79 (dd, J = 13.0, 3.7 Hz, 1H), 3.57 (dd, J = 7.2, 3.6 Hz, 1H), 3.20 (dd, J = 12.9, 4.8 Hz, 1H), 2.38 (s, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 161.31, 157.19, 156.84, 156.41, 136.55, 136.22, 135.38, 131.94, 128.55, 126.44, 124.00, 123.31, 114.87, 110.31, 68.15, 54.80, 45.24, 42.93, 19.75. IR (ATR): ν 3298, 2923, 1545, 1330, 1061, 742 cm^{-1}. $[\alpha]_D$ = –44.5 (c 0.20, CH_3OH). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{18}H_{16}ClF_3N_2O_3$ + H^+]: 401.0874, found 401.0885.

(S)-2-(3-Chloro-4-methylphenyl)-2-((S)-6,7-dibromo-1-oxo-1,2,3,4-tetrahydropyrrolo[1,2-a]pyrazin-4-yl)ethyl 2,2,2-Trifluoroacetate(15): The title compound was prepared following the procedure for **10** starting from **14** (25 mg, 0.062 mmol) and NBS (23 mg, 0.124 mmol) to afford **15** in a 75 % yield (26 mg, 0.046 mmol) as a yellow oil. TLC: R_f 0.35 (Et_2O = 100 %). ^1H NMR (400 MHz, $CDCl_3$) δ = 7.30–7.24 (m, 2H), 7.11–7.05 (m, 2H), 6.19 (d, J = 4.9 Hz, 1H), 4.63 (dd, J = 10.5, 2.7 Hz, 1H), 4.53 (dd, J = 11.3, 7.0 Hz, 1H), 4.26 (dd, J = 11.3, 5.4 Hz, 1H), 3.72 (dd, J = 13.2, 3.6 Hz, 1H), 3.67–3.58 (m, 1H), 3.23 (dd, J = 13.2, 4.5 Hz, 1H), 2.39 (s, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 158.76, 156.95, 156.52, 136.92, 135.51, 135.41, 132.03, 128.42, 126.36, 125.53, 125.12, 117.19, 107.74, 101.69, 67.53, 55.19, 46.09, 43.02, 19.80. IR (ATR): $\tilde{\nu}$ = 3234, 1775, 1705, 1335, 1181, 749 cm^{-1}. $[\alpha]_D$ = –6.0 (c 0.10, CH_3OH). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{18}H_{14}Br_2ClF_3N_2O_3$ + H^+]: 556.9085, found 556.9075.

(S)-6,7-Dibromo-4-((S)-1-(3-chloro-4-methylphenyl)-2-hydroxyethyl)-3,4-dihydropyrrolo[1,2-a]pyrazin-1(2H)-one (16): The title compound was prepared following the procedure for **11** starting from **15** (15 mg, 0.029 mmol) and NaOH (0.1 mL, 1 M) to afford **16** in a 99 % yield (13.2 mg, 0.029 mmol) as a yellow oil. TLC: R_f 0.40 (EtOAc = 100 %). ^1H NMR (400 MHz, $CDCl_3$) δ = 7.29 (d, J = 1.3 Hz, 1H), 7.26–7.22 (m, 1H), 7.12–7.05 (m, 1H), 7.03 (s, 1H), 6.29 (d, J = 5.0 Hz, 1H), 4.52 (dd, J = 10.2, 3.0 Hz, 1H), 3.88 (dd, J = 10.6, 8.9 Hz, 1H), 3.67 (dd, J = 13.2, 3.7 Hz, 1H), 3.50 (dd, J = 10.8, 4.6 Hz, 1H), 3.42–3.32 (m, 1H), 3.23 (dd, J = 13.1, 5.3 Hz, 1H), 2.37 (s, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ = 158.90, 137.55, 136.12, 135.29, 131.83, 128.65, 126.80, 125.20, 116.60, 108.05, 101.10, 63.35, 55.59, 49.73, 42.94, 19.77. IR (ATR): $\tilde{\nu}$ = 3310, 2879, 1660, 1335, 1053, 749 cm^{-1}. $[\alpha]_D$ = +12.5 (c 0.20, CH_3OH). HRMS (ESI/Q-TOF) m/z calcd. for [$C_{16}H_{15}Br_2ClN_2O_2$ + H^+]: 460.9262, found 460.9276.

2-(3-Chloro-4-methylphenyl)-2-((S)-6,7-dibromo-1-oxo-1,2,3,4-tetrahydropyrrolo[1,2-a]pyrazin-4-yl)acetic Acid (17): The title compound was prepared following the procedure for **Longamide B** starting from **16** (4.00 mg, 0.009 mmol), BAIB (7.24 mg, 0.022 mmol), $NaHCO_3$ (3.00 mg, 0.036 mmol) and TEMPO (0.15 mg, 0.001 mmol) to afford a crude product that was purified by column chromatography to afford **17** as an inseparable mixture of diastereoisomers (44 %:56 %) in a 83 % yield (3.56 mg, 0.007 mmol) as a pale yellow amorphous solid. R_f 0.50 (EtOAc = 100 %). ^1H NMR (400 MHz, $CDCl_3$) mixture of diastereomers (44 %:56 %), δ = 7.42 (bs, 1H), 7.19 (s, 0.5H), 7.16–7.09 (m, 1.5H), 7.01–6.85 (m, 2H), 4.86–4.78 (m, 1H), 4.03 (d, J = 6.1 Hz, 1H), 3.93–3.77 (m, 1.5H), 3.76–3.72 (m, 0.5H), 2.34–2.27 (m, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) mixture of diastereomers (44 %:56 %) δ = 159.96, 159.18, 136.73, 134.86, 132.07, 131.43, 131.31, 129.42, 127.35, 125.03, 116.51, 108.65, 101.68, 55.79, 55.00, 53.19, 42.39, 29.70, 19.76. IR (ATR): $\tilde{\nu}$ = 3255,

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2923, 2498, 2161, 1201, 751 cm^{-1}. HRMS (ESI/Q-TOF) m/z calcd. for $[C_{16}H_{13}Br_2ClN_2O_3 + H^+]$: 474.9054, found 474.9068.

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